1. **Describe the steps in development of hydropower in Nepal with basic characteristics. Name the hydropower development institution in Nepal.**

Answer:

Hydropower development is a systematic process starting from identification of a potential site and ending with operation and maintenance of the power plant. In Nepal, the process generally follows these major steps:

i. **Identification of Hydropower Potential**

- Identify rivers and potential sites based on river discharge, available head, topography and geology.
- Preliminary studies and hydropower potential mapping are carried out.

ii. **Reconnaissance and Pre-feasibility Study**

- Conduct a preliminary field investigation of the proposed site.
- Study topography, geology, hydrology, accessibility, transmission possibilities and environmental conditions.
- Estimate the approximate installed capacity, energy generation and project cost.

iii. **Feasibility Study**

A detailed feasibility study is conducted to establish whether the project is technically and economically viable. It includes:

- Detailed hydrological analysis and design discharge.
- Geological and geotechnical investigations.
- Topographical survey.
- Selection of dam/intake, headrace, tunnel, penstock and powerhouse locations.
- Hydraulic and structural design.
- Environmental and social assessment.
- Cost estimate, financial analysis and economic evaluation.

iv. **Detailed Project Design**

After establishing feasibility, the project is designed in detail, including:

- **Headworks:** dam/weir, intake, gravel trap, settling basin/desander.
- **Water conveyance:** canal, tunnel, forebay and penstock.
- **Powerhouse:** turbine, generator and auxiliary systems.
- Tailrace and river outlet.
- Detailed drawings, specifications, bill of quantities and construction schedule are prepared.

v. **Licensing and Approvals**

The developer obtains the necessary approvals from concerned authorities. In Nepal, important regulatory processes include:

- Survey and generation licensing under the prevailing electricity laws.
- Environmental approval such as IEE/EIA, depending on project size and applicable regulations.
- Land acquisition and forest-related approvals where required.

vi. **Financial Closure and Contracting**

- Total project cost and financing requirements are finalized.

- Financing may come from equity, commercial loans, development banks, government investment or other sources.
- EPC, civil works, electro-mechanical and other construction contracts are awarded.

vii. **Construction**

The physical construction of the project is carried out in stages:

- Access roads, camps and temporary facilities.
- Diversion works and cofferdams.
- Dam/weir and intake.
- Tunnels, canals and other water-conveyance structures.
- Penstock and powerhouse.
- Electromechanical equipment installation.

viii. **Testing and Commissioning**

- Individual equipment and systems are tested.
- Turbine-generator units undergo trial operation.
- Hydraulic, electrical, protection and control systems are checked.

ix. **Operation and Maintenance**

During operation:

- Water resources and reservoir/intake systems are managed.
- Turbines, generators, electrical equipment and civil structures are regularly inspected.
- Sediment, floods, landslides and GLOF risks are monitored, which are particularly important for Himalayan projects.
- Preventive and corrective maintenance is performed to maintain reliable generation.

x. **Rehabilitation and Upgrading**

With time, equipment and structures may require:

- Major maintenance and replacement.
- Turbine rehabilitation due to sediment erosion.
- Capacity upgrading and efficiency improvement.
- Safety strengthening and modernization of control systems.

Hydropower development in Nepal involves government ministries, regulatory agencies, public utilities, investment institutions, private developers, and supporting organizations. The different hydropower development institution in Nepal is listed below:

i. **Policy-making and Government Institutions**

a. Ministry of Energy, Water Resources and Irrigation (MoEWRI)
b. National Planning Commission (NPC)
c. Ministry of Finance (MoF)

ii. **Regulatory and Licensing Institutions**

a. Department of Electricity Development (DoED)
b. Water Resources Planning and Technical Institutions
c. Water and Energy Commission Secretariat (WECS)
d. Department of Hydrology and Meteorology (DHM)
e. Water Resources Research and Development Centre (WRRDC)

iii. **Electricity Generation, Transmission and Distribution Institutions**

a. Nepal Electricity Authority (NEA)
b. Vidhyut Utpadan Company Limited (VUCL)

c. Rastriya Prasaran Grid Company Limited (RPGCL)

iv. **Hydropower Investment and Financing Institutions**
   a. Hydroelectricity Investment and Development Company Limited (HIDCL)
   b. Commercial Banks and Financial Institutions

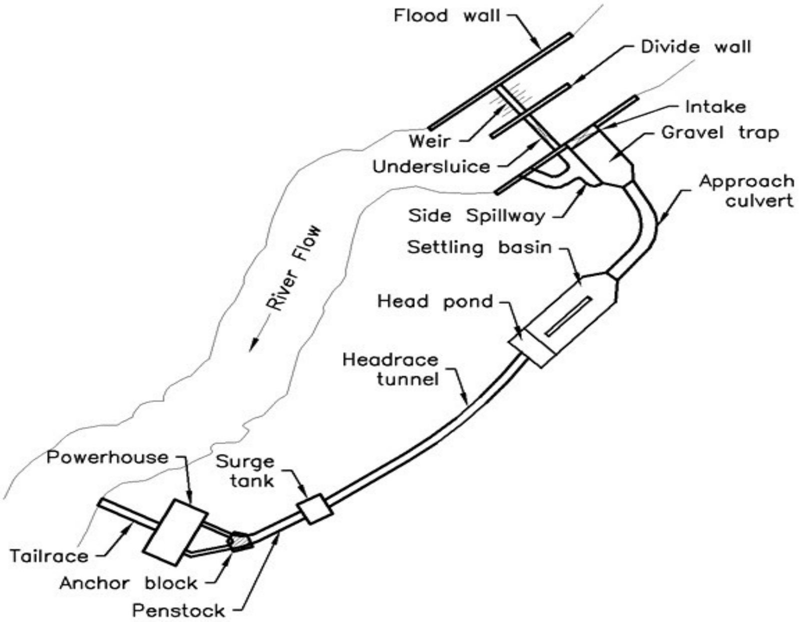
v. **Private Sector and Independent Power Producers (IPPs)**

vi. **Environmental and Social Institutions**
   a. Ministry of Forests and Environment (MoFE) and its concerned agencies:
   b. Local Governments (Municipalities and Rural Municipalities)

2. **Draw the typical layout of hydropower house showing all essential components.**

   Answer:

   The typical layout and major components of a typical run-of-river hydropower plant are given below:



Components:

i. **Weir:**
   - ➢ It is constructed across the river to divert the part of river flow to various water conveyance systems through the intake.

ii. **Intake:**
   - ➢ It is a structure constricted to divert the required amount of water into a conduit leading to the power plant.
   - ➢ Trash racks are provided to prevent entry of floating debris and coarse bed load into water conveyance systems.

iii. **Gravel trap:**
   - ➢ It is required to flush out bed sediments that enter the approach canal back into a river.
   - ➢ The gravel trap is constructed close to the intake in order to prevent gravel from getting into the approach channel.

iv. **Settling basin:**
   - ➢ It is the structure to remove suspended sediments form the conveyance water for power plant.
   - ➢ It shall be designed to ensure that the water entering the water conveyance system is free from sediments that can damage the penstock and turbine runners.

v. **Headrace conveyance:**

- A canal or pipe or tunnel or combination of these are provided for delivering the design flow to the forebay or surge tank with a minimum energy loss.

vi. **Forebay:**
- It is located at the beginning of penstock shaft that supplies required flow to the turbine during start up, accommodate the reject flow during shut down, reduce water hammer effect.
- It serves as a secondary settling basin as well as a sediment sluice to remove settle suspended particles.

vii. **Surge tank:**
- It is located between the headrace pressure conduits and the steeply sloping penstock pipe.
- It is generally a cylindrical storage reservoir connected to penstock pipe close to turbine.

viii. **Penstock:**
- It is a pipe that conveys the flow from forebay or surge tank to the turbine.
- They are designed to carry water to turbines with the least possible loss of head consistent with the overall economy of the project.

ix. **Anchor blocks:**
- It is an encasement of a penstock designed to restrain the pipe movement in all durations.
- Anchor block should be placed at all sharp horizontal and vertical bends, as there are forces at such bends which tends to move the pipe out of the designated alignment.

x. **Powerhouse:**
- It is a structure complex where all the equipment for providing electricity is suitably arranged.
- Two requirements of powerhouse planning are functional efficiency and aesthetic beauty.

xi. **Tailrace:**
- It disposes the design flow discharged by hydraulic turbine after the power generation.

3. **What are the major factor needed to be considered while optimizing the dam height? Explain briefly.**

Answer:

Optimization of dam height means determining the most economical and technically suitable dam height that provides the required benefits without causing excessive construction and associated costs.

The major factors that needed to be considered while optimizing the dam height are as follows:

i. **Storage requirement**

a. The dam height should provide sufficient reservoir storage for the required firm power, irrigation, water supply and flood control.

b. Higher the dam, larger the storage, but the benefit eventually becomes marginal.

ii. **Hydropower potential**

a. For hydropower, increased dam height increases the gross head and storage, potentially increasing energy generation.

b. The optimum height is reached when the additional energy benefit no longer justifies the additional cost.

iii. **Topography and valley shape**

a. The shape and width of the valley strongly influence reservoir volume and dam volume.

b. A narrow gorge may allow a higher dam with relatively smaller dam volume, while a wide valley can make additional height expensive.

iv. **Geological and geotechnical conditions**

a. Foundation strength, permeability, faults, joints, landslide-prone slopes and abutment stability must be considered.

b. Poor geology may limit the permissible dam height or require expensive foundation treatment.

v. **Flood and spillway requirements**
   a. The dam must safely pass the design flood and probable maximum flood (where applicable).
   b. Increasing dam height also affects reservoir routing, freeboard and spillway requirements.

vi. **Sedimentation**
   a. Adequate dead storage must be provided to accommodate sediment deposition during the project life.
   b. In sediment-laden Himalayan rivers, this is particularly important.

vii. **Submergence and resettlement**
   a. Higher dam height increases the area submerged, potentially affecting:
      i. Settlements and population
      ii. Agricultural land
      iii. Forests
      iv. Roads and bridges
      v. Cultural and historical sites

viii. **Environmental and social impacts**
   a. Impacts on aquatic ecosystems, downstream flow, biodiversity and local communities increase with reservoir size.
   b. Environmental mitigation and compensation costs should be included.

ix. **Construction cost**
   a. Cost of the dam body, foundation treatment, diversion works, spillway, outlets and associated structures increases with height.
   b. The optimum height should minimize the cost per unit of benefit/energy.

4. **How does a daily pondage basin in a PRoR project help to regulate the fluctuation in demand? Explain.**

Answer:

A daily pondage basin is a small storage reservoir provided in a pondage-type run-of-river (PRoR) hydropower project to store water during periods of low electricity demand and release it during periods of high demand.

In a conventional RoR project, power generation directly follows river discharge, so the plant has limited ability to respond to changes in electricity demand. A daily pondage basin provides short-term storage and separates the timing of water inflow from power generation.

- **During low-demand hours**, such as late night, the demand for electricity is low. The plant operates at reduced output, while the incoming river water is stored in the pondage basin.
- **During peak-demand hours**, the stored water is released through the water conductor to the turbines. The plant can therefore operate at higher output than the instantaneous river inflow would permit.

This process is repeated approximately on a daily cycle, hence the term daily pondage. It allows the plant to follow the daily load curve more effectively and reduces unnecessary spilling of water during low-demand periods. It also improves the capacity utilization and peaking capability of a PRoR plant without requiring a large seasonal reservoir.

Suppose river inflow is relatively constant at 100 m^3/s, but electricity demand varies during the day:

- At Night: let's suppose only 60 m^3/s is required for generation due to less demand. So, 40 m^3/s is stored.
- During Peak hours: let's suppose140 m^3/s is required. The additional 40 m^3/s comes from pondage storage.

5. **How do you fix duration of construction of a hydropower project? Explain the concept of CPM in project planning?**

Answer:
The construction duration of a hydropower project is fixed by determining the time required to complete all major construction activities, considering their sequence, interdependence, available resources, site conditions, seasonal constraints and construction methods.

The following procedure is generally followed:

i. **Break the project into major activities**
Divide the project into activities such as access road, camp establishment, river diversion, headworks, tunnels, surge tank, penstock, powerhouse, transmission line, electromechanical installation, testing and commissioning.

ii. **Estimate quantities of each activity**
Determine the volume or quantity of excavation, concrete, reinforcement, tunnelling, lining, installation, etc.

iii. **Determine productivity**
Estimate the daily or monthly output of available manpower, machinery and equipment.

iv. **Calculate duration of individual activities**

$$\text{Duration} = \frac{\text{Quantity of work}}{\text{Productivity}}$$

Appropriate allowances are made for mobilization, equipment maintenance and other interruptions.

v. **Establish logical sequence and dependencies**
Determine which activities must precede or follow others and which activities can be carried out simultaneously.

vi. **Consider site and seasonal conditions**
For Nepalese hydropower projects, monsoon floods, landslides, difficult Himalayan geology, river diversion periods, snowfall and access constraints can significantly affect the schedule.

vii. **Prepare the project network/schedule**
Techniques such as CPM (Critical Path Method) or PERT can be used to establish the overall project schedule and identify critical activities.

viii. **Determine the critical path**
The longest sequence of dependent activities determines the minimum practical construction period. Delay in critical activities can delay the whole project.

ix. **Optimize the duration**
If necessary, construction time can be reduced by increasing resources, using additional working faces, multiple shifts, improved equipment or better construction methods.

x. **Provide reasonable contingency**
Additional time is allowed for unforeseen geological conditions, floods, supply delays, equipment breakdown, labor problems and other risks.

CPM (Critical Path Method) is a network-based, planning and scheduling technique used to plan, schedule, control and monitor project activities. It is particularly useful in large construction projects such as hydropower projects, dams, tunnels and powerhouses, where many activities are interdependent.

In CPM, a project is divided into a number of activities, and the logical relationship between these activities is represented by a network diagram. For each activity, its duration is estimated and the network is analyzed to determine:

i. Earliest Start (ES)
ii. Earliest Finish (EF)
iii. Latest Start (LS)
iv. Latest Finish (LF)
v. Float/Slack
vi. Critical Path

The critical path is the longest-duration path through the project network. It determines the minimum possible project completion time. Activities lying on the critical path generally have zero total float. Therefore, any delay in a critical activity directly delays the completion of the entire project unless corrective action is taken.

The main steps involved in CPM:

i. List all project activities.
ii. Determine the logical sequence/dependency between activities.
iii. Estimate the duration of each activity.
iv. Prepare the project network diagram.
v. Perform forward pass to calculate ES and EF.

$$EF = ES + Duration$$

vi. Perform backward pass to calculate LS and LF.

$$LS = LF - Duration$$

vii. Calculate float for each activity.

$$TF = LS - ES = LF - EF$$

viii. Identify the critical path.
ix. Determine the minimum project completion time.
x. Monitor progress and take corrective measures if critical activities are delayed.

Importance of CPM:

CPM helps the project manager to:

- Determine the minimum project duration.
- Identify activities that require close monitoring.
- Allocate labor, equipment and financial resources efficiently.
- Identify activities having time flexibility (float).
- Assess the effect of delays on project completion.
- Improve project scheduling and coordination.
- Assist in time–cost optimization, such as deciding where crashing can economically reduce project duration.

6. **Explain the safety issues and measures to be taken into consideration while constructing a tunnel?**

Answer:

The process of constructing tunnel is a high-risk underground activity as workers are exposed to unstable rock, falling debris, blasting, groundwater, poor ventilation, confined spaces, machinery and sudden changes in geological conditions.

The recent catastrophic flood in Rasuwa clearly demonstrates why strict safety rules and disaster preparedness are essential in hydropower construction. The sudden flooding severely affected hydropower projects and infrastructure along the Bhotekoshi–Trishuli river corridor. The mud water flooded the entire tunnel where hundreds of workers have gone missing. This disaster highlighted the serious risks faced by workers at hydropower construction sites. Rescue operations focused on workers affected or trapped at project sites and tunnels are underway. Hence, the reliable emergency communication, evacuation procedures, rescue access, and emergency preparedness is very essential.

The safety issues while constructing tunnel are listed below:

i. Unstable rock masses and roof collapse
ii. Faults, joints and weak zones
iii. Rock falls and landslides at portals

iv. Sudden changes in geological conditions
v. Excessive vibration and overbreak
vi. Toxic blasting fumes
vii. Misfired explosives
viii. Sudden groundwater inflow
ix. High-pressure water in fractured rock
x. Flooding of the tunnel
xi. Erosion or weakening of tunnel support
xii. Oxygen deficiency
xiii. Accumulation of blasting fumes and dust

Therefore, safety must be considered at every stage of excavation and construction. The following safety measures should be considered inside the underground power house:

i. **Adequate ventilation**
   a. Provide mechanical ventilation to supply fresh air and remove heat, smoke, dust and harmful gases.
   b. Monitor oxygen and toxic-gas levels where necessary.

ii. **Fire protection**
   a. Install fire alarms, smoke detectors, fire extinguishers and suitable fire-fighting systems.
   b. Provide fire-resistant doors and clearly marked emergency exits.
   c. Maintain separate and safe storage of flammable materials.

iii. **Electrical safety**
   a. Properly earth all electrical equipment.
   b. Provide insulation, circuit breakers, protective relays and emergency shutdown systems.
   c. Restrict access to high-voltage areas to authorized personnel only.

iv. **Protection against water**
   a. Provide effective drainage, sump pumps and standby pumping arrangements.
   b. Monitor seepage and water pressure, particularly around the powerhouse cavern and tunnels.

v. **Rock and structural stability**
   a. Ensure proper rock bolting, shotcrete, lining and other support systems.
   b. Regularly inspect the roof, walls and support structures for cracks, loose rocks and deformation.

vi. **Safe access and escape**
   a. Provide properly illuminated stairways, walkways and emergency exits.
   b. Keep escape routes free from obstruction and clearly mark them with signs.

vii. **Lighting and communication**
   a. Provide adequate normal and emergency lighting.
   b. Install reliable telephone/radio communication with the control room and outside emergency services.

viii. **Machinery and moving equipment safety**
   a. Guard rotating and moving parts of turbines, generators and auxiliary machinery.
   b. Provide emergency-stop devices and maintain safe working distances.

ix. **Personal protective equipment (PPE)**
   a. Workers should use helmets, safety shoes, gloves, eye/ear protection and other PPE appropriate to the work.

x. **Emergency preparedness**
   a. Prepare emergency procedures for fire, flooding, electrical accidents, rockfall and equipment failure.
   b. Conduct regular safety drills and ensure workers know evacuation procedures.

7. **What is a risk analysis? How do you carry out risk analysis for a hydropower project? Name five major potential risks in the construction of a hydropower project.**

Answer:

Risk analysis is the systematic process of identifying, assessing, and quantifying uncertainties and potential adverse events that may affect the cost, schedule, safety, quality, environmental performance, and successful completion of a hydropower project. It helps the project team take appropriate measures to reduce the probability and/or impact of risks.

Procedure for Risk Analysis of a Hydropower Project

Risk analysis can generally be carried out through the following steps:

i. **Risk identification:** Identify all possible risks from geological, hydrological, technical, financial, environmental, social, contractual and construction conditions.
ii. **Risk classification:** Group the identified risks according to their nature, such as technical, natural, financial, environmental and contractual risks.
iii. **Risk assessment:** Estimate the probability of occurrence and the potential consequences/impact of each risk.
iv. **Risk rating and prioritization:** Rank risks using a risk matrix, generally combining probability and impact:

$$\text{Risk Rating} = \text{Probability} \times \text{Impact}$$

v. **Quantitative analysis:** For major risks, estimate their effects on project cost, completion time and expected energy generation using methods such as sensitivity analysis, scenario analysis and Monte Carlo simulation.
vi. **Risk response:** Develop measures to avoid, reduce, transfer or accept each significant risk. For example, additional geological investigation may reduce geological uncertainty, while insurance can transfer certain risks.
vii. **Contingency planning:** Provide appropriate time and cost contingencies for significant residual risks.
viii. **Monitoring and review:** Continuously monitor identified risks and identify new risks throughout investigation, design, construction and operation.

The five major potential risks in the construction of a hydropower project are listed below:

i. **Geological and geotechnical risk**: unexpected faults, weak rock, unstable slopes, landslides, excessive seepage and difficult tunnelling conditions.
ii. **Hydrological and flood risk**: floods, GLOFs, extreme rainfall, sediment-laden floods and unexpected variation in river discharge.
iii. **Construction and technical risk**: tunnel collapse, construction accidents, equipment failure, design deficiencies and difficulties in underground works.
iv. **Cost and schedule risk**: inflation, exchange-rate fluctuations, shortage of construction materials, contractor problems, delays in procurement and unforeseen construction conditions.
v. **Environmental and social risk**: land acquisition, resettlement, impacts on aquatic ecosystems and downstream water users, forest clearance and conflicts with local communities.

8. **Define specific speed, synchronous speed, runway speed, speed factor and design speed of turbine.**

Answer:

i. Specific Speed (N_s):

Specific speed of a turbine is the speed at which a geometrically similar turbine would run if it developed unit power under unit head.

It is given by:

$$N_s = \frac{N\sqrt{P}}{H^{5/4}}$$

$$where,$$

$$N = turbine \ \ speed\left(rpm\right)$$

$$P = power \ \ developed\left(kW\right)$$

$$H = net \ \ head\left(m\right)$$

Specific speed is useful for selecting the appropriate type of turbine.

ii. Synchronous Speed (Ns)

Synchronous speed is the speed at which the magnetic field of an AC generator rotates. The turbine-generator must run at this speed when directly coupled.

$$N = \frac{120f}{p}$$

$$where,$$

$$f = frequencyofelectricalsupply\left(Hz\right)$$

$$p \ = \ number \ \ of \ \ generator \ \ poles$$

$$For \ \ Nepal's \ \ grid \ \ frequency \ \ of \ \ 50\,Hz:$$

$$N = \frac{6000}{p}\,\text{rpm}$$

iii. Runaway Speed / Runway Speed

Runaway speed is the maximum rotational speed attained by a turbine when it is operating at full load and the generator is suddenly disconnected from the electrical system, while the turbine continues to receive water.

It is important in designing the turbine shaft, runner, bearings and other rotating components.

Generally,

$$N_{\text{runaway}} > N_{\text{rated}}$$

The exact runaway speed depends on the turbine type and operating conditions.

iv. Speed Factor (Ku)

The speed factor is the ratio of the peripheral speed of the turbine runner at its characteristic diameter to the theoretical velocity of water corresponding to the available head.

$$K_u = \frac{u}{\sqrt{2gH}}$$

$$where,$$

$$u = peripheral \ \ speed \ \ of \ \ runner\left(m/s\right)$$

$$H = net \ \ head\left(m\right)$$

$$g = acceleration\ due\ ¿\ gravity.$$

It is useful in turbine selection and runner-diameter determination.

v. Design Speed

Design speed is the rotational speed at which the turbine is designed to operate under its rated/design head and discharge while producing the specified output.

It is normally selected to be compatible with the synchronous speed of the generator in a directly coupled turbine-generator unit.

Thus,

$$N_{\text{design}} \approx N_{\text{synchronous}}$$

for a directly coupled unit.

9. **Discuss the different issues and challenges associated with the sediment problem in hydropower project in Nepal. Explain how do you design run-off river headworks of a hydropower project to overcome such sediment problem in sediment laden river. Explain the feature of good headwork at Himalayan River.**

Answer:

Sediment is one of the major challenges in Nepalese hydropower development. Himalayan rivers originate in young, rapidly uplifting mountains and carry large quantities of suspended and bed-load sediment, especially during the monsoon. The problem becomes more severe because of landslides, debris flows, glacial activity and unstable slopes.

The different issues and challenges associated with the sediment problem in hydropower project are listed below:

a. **High sediment concentration**
Nepalese rivers can carry very high sediment concentrations during the monsoon. Fine silt, sand, gravel and occasionally larger boulders enter the hydropower system. The high sediment load reduces the effective life and efficiency of hydraulic structures.

b. **Abrasion of turbines**
This is one of the most serious problems. Hard particles such as quartz and other abrasive minerals carried by Himalayan rivers strike turbine components at high velocity. It causes erosion of runner blades, damage to guide vanes and nozzles, reduction in turbine efficiency etc. which may require frequent turbine repair and replacement.

c. **Heavy flushing requirement**

To remove deposited sediment, projects may require sediment flushing through sluices, under sluices or flushing tunnels. However, flushing can be difficult where the reservoir geometry does not permit effective flushing.

d. **Landslides and debris flows**

Nepal's steep and geologically young terrain is highly susceptible to landslides, rockfalls and debris flows. These can suddenly introduce large quantities of sediment and coarse material into rivers.

A major landslide upstream can therefore create a temporary but severe sediment problem for a hydropower project.

e. **Glacial and GLOF-related sediment**

In high Himalayan catchments, glaciers and glacial lakes can contribute substantial quantities of sediment. A Glacial Lake Outburst Flood (GLOF) may transport enormous amounts of sediment, boulders, trees and debris downstream.

In Nepal, run-of-river (RoR) hydropower projects are commonly located on steep Himalayan rivers having high sediment concentration, abrasive particles, large seasonal variation in discharge, and risk of landslides/debris flows. Therefore, the headworks should be designed not only to divert water but also to exclude, settle, flush and safely pass sediment before it reaches the turbines.

Run-off river headworks of a hydropower project is designed as following to overcome such sediment problem in sediment laden river:

i. **Basic arrangement of sediment-conscious RoR headworks**
A typical arrangement is:
River → Diversion weir/barrage → Undersluice → Intake → Gravel trap/desander → Headrace tunnel → Forebay/surge arrangement → Penstock → Turbine
The headworks should provide separate paths for clean water to the power system and sediment-laden water back to the river.

ii. **Selection of intake location**
The intake should be located where the river is relatively stable and the flow is reasonably uniform.
The intake should preferably be:
   a. Away from the direct deposition zone.
   b. Located on the outer bank of a suitable bend, where excessive sediment deposition is less likely.
   c. Protected from debris and bank erosion.
   d. Positioned so that the approach flow is smooth and free from excessive turbulence.
A proper approach channel and intake orientation are essential because poor intake geometry can draw sediment directly into the power tunnel.

iii. **Provision of a diversion weir/barrage**
A low diversion weir or barrage raises the river water level sufficiently to divert the required discharge into the intake. The structure should not unnecessarily trap large quantities of sediment.

iv. **Undersluice**
Undersluices are particularly important in sediment-laden Himalayan rivers**.** The undersluice gates should be capable of rapid operation during high sediment concentration and flood events.

v. **Intake design for sediment exclusion**
The intake should be designed so that the amount of sediment entering the headrace is minimized. The intake sill should be sufficiently elevated above the river bed to reduce the entry of coarse bed load while still allowing the required design discharge to enter.

vi. **Gravel trap / settling arrangement**
Coarse sediment such as gravel and coarse sand should be removed before water enters the main desanding basin or headrace tunnel. A gravel trap may be provided immediately downstream of the intake.

vii. **Desanding basin**
The desanding basin is the principal component for protecting turbines from abrasive sediment. Its purpose is to settle particles above a specified size before the water enters the headrace.
The basin generally consists of:
Inlet transition → Settling chamber → Sediment collection zone → Flushing arrangement → Outlet transition

viii. **Efficient flushing system**
Because sediment accumulation is unavoidable, the desander should have an efficient flushing system. The system should be capable of removing accumulated sediment without excessive interruption of power generation.

ix. **Provision for sediment monitoring**

A modern RoR headworks should include regular monitoring of bed load, suspended sediment concentration, particle size distribution, turbidity etc. During periods of exceptionally high sediment concentration, the intake may be temporarily closed or the plant operated under a suitable sediment-management protocol.

The features of a good headwork in a Himalayan River are listed below:

i. The headworks should preferably be founded on the stable strata as Himalayan rivers have high flood velocities and severe scour potential.
ii. A good headwork should allow coarse sediment to pass downstream and provide efficient settling and flushing of finer sediment.
iii. Sediment deposited around the intake and within the desander should be removable easily and quickly.
iv. The headworks must safely pass extreme floods without causing unacceptable afflux, erosion or structural damage.
v. The structures should be robust and capable of safely handling sudden increases in discharge and sediment/debris load.
vi. Gates, trash racks, desanders and flushing facilities should be accessible for inspection, cleaning, repair and emergency operation, particularly during the monsoon.
vii. The headworks should achieve sediment exclusion without causing excessive hydraulic losses, because unnecessary head loss directly reduces the available energy for power generation.
viii. A good Himalayan headwork should be simple, robust and economical, with as few vulnerable mechanical components as practical.

10. **Describe briefly about different types of field investigation required for hydropower project.**
Answer:
Field investigation is the systematic collection and examination of physical, geological, hydrological, environmental, and socio-economic information at the proposed hydropower project site. It is carried out through site visits, surveys, measurements, mapping, sampling, and testing to determine the feasibility and suitability of the project.

The different types of field investigation required for hydropower projects are as follows:

a. Topographical Investigation

- Preparation of detailed contour maps.
- Longitudinal and cross-sectional surveys of rivers.
- Survey of dam, intake, powerhouse, tunnel and canal sites.
- Survey for access roads, transmission lines and construction facilities.

b. Hydrological Investigation

- Measurement of river discharge and water levels.
- Collection of rainfall and climatic data.
- Flood-frequency analysis.
- Assessment of sediment load and water availability.

c. Geological Investigation

- Geological mapping of the project area.
- Identification of rock types, faults, folds, joints and discontinuities.
- Geological investigation along tunnels and pressure waterways.
- Identification of landslide and unstable zones.

d. Geotechnical Investigation

- Collection of soil and rock samples.
- Determination of bearing capacity and shear strength.

- Permeability and groundwater investigation.
- Rock-mass classification and foundation assessment.

e. Seismic Investigation

- Study of regional seismicity and active faults.
- Assessment of earthquake hazards.
- Determination of appropriate seismic design parameters.

11. **a. Define Froude Number**

b. Estimate the tail water elevation required to form a hydraulic jump from the following data:

- **spillway elevation= 150 m**
- **elevation of energy line just U/S od spillway crest= 152.5 m**
- **elevation of horizontal apron= 100 m**
- **coefficient of discharge = 0.68**
- **neglect the energy loss due to flow over spillway.**

Answer:

a. The Froude number is the ratio of inertial force to gravitational force**.**

$$Fr = \frac{\text{Inertial Force}}{\text{Gravitational Force}}$$

For open-channel flow:

$$Fr = \frac{V}{\sqrt{gL}}$$

For a rectangular open channel:

$$Fr = \frac{V}{\sqrt{gD}}$$

$$where:$$

$$V = \ mean\ \ velocity\ \ of\ \ flow$$

$$g = \ acceleration\ \ due\ \ \text{¿}\ \ gravity$$

$$L = \ characteristic\ \ length$$

$$D\ =\ hydraulic\ \ depth$$

Significance of different Froude number values

Froude Number	Type of Flow
Fr<1	Subcritical flow
Fr=1	Critical flow
Fr>1	Supercritical flow

Fr<1: Gravity forces dominate. Surface disturbances can travel upstream.

Fr=1: Flow is at the critical condition.

Fr>1: Inertial forces dominate. Surface disturbances cannot travel upstream.

Significance: Froude number is important in the design of spillways, canals, rivers, hydraulic jumps, and other open-channel structures.

Given data

$$Spillway\ \ crest\ \ elevation, Z_c = 150\,\mathrm{m}$$

$$Energy\ \ line\ \ just\ \ upstream\ \ of\ \ crest, E_u = 152.5\,\mathrm{m}$$

$$Apron\ \ elevation, Z_a = 100\,\mathrm{m}$$

$$Coefficient\ \ of\ \ discharge, C_d = 0.68$$

$$g = 9.81\,\mathrm{m/s^2}$$

$$Energy\ \ loss - spillway\ \ is\ \ neglected.$$

We need to find the tail-water elevation required to form a hydraulic jump.

Head over spillway crest :

$$H = 152.5 - 150 = 2.5\,\mathrm{m}$$

$$For\ \ a\ \ spillway,\ \ discharge\ \ per\ \ unit\ \ width\ \ is$$

$$q = C_d \frac{2}{3}\sqrt{2g}\,H^{3/2}$$

$$Substituting,$$

$$q = 0.68 \times \frac{2}{3}\sqrt{2(9.81)}(2.5)^{3/2}$$
$$q = 7.94\,\mathrm{m^3/s\ per\ m\ width}$$

Since energy loss over the spillway is neglected, the total energy at the apron is

$$E = 152.5 - 100 = 52.5\,\mathrm{m}$$

Therefore,

$$52.5 = y_1 + \frac{V_1^2}{2g}$$

$$Since$$

$$V_1 = \frac{q}{y_1}$$

$$we\ \ get$$

$$52.5 = y_1 + \frac{q^2}{2g\,y_1^2}$$

$$\textit{Substituting}\quad q = 7.94,$$

$$52.5 = y_1 + \frac{(7.94)^2}{2(9.81)\,y_1^2}$$

$$\textit{Solving},$$

$$y_1 \approx 0.248\,\mathrm{m}$$
$$V_1 = \frac{7.94}{0.248}$$
$$V_1 \approx 32.02\,\mathrm{m/s}$$
$$Fr_1 = \frac{V_1}{\sqrt{g\,y_1}}$$
$$Fr_1 = \frac{32.02}{\sqrt{9.81(0.248)}}$$
$$Fr_1 \approx 20.53$$

This indicates a highly supercritical flow.

For a hydraulic jump in a rectangular channel,

$$\frac{y_2}{y_1} = \frac{1}{2}\left[\sqrt{1 + 8Fr_1^2} - 1\right]$$

$$\textit{Thus},$$

$$y_2 = \frac{0.248}{2}\left[\sqrt{1 + 8(20.53)^2} - 1\right]$$
$$y_2 \approx 7.08\,\mathrm{m}$$

The apron elevation is 100 m. Therefore,
Tail water elevation= $100 + y_2 = 100 + 7.08$

$$\text{Tail-water elevation} \approx 107.08\,\mathrm{m}$$